

POTHOLELOG · DESIGN DECISION RECORD

PotholeLog TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	PHL-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potholelog		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 (first pilot partner) decided by Amish on 2026-10-02
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 work and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** items D1 to D7 decided by Amish, 2026-09-25: go with recommendation (see PHL-DDR-002); item O1 decided by Amish on 2026-10-02 (PHL-DEC-001).

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed eight items as "Proposed, awaiting Amish", seven of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Under that instruction, each item that carries a recommendation is adopted as recommended for TRL 3 work and stays open for his review. The item without a specific recommendation stays open. Nothing in v0.1 of this record was a decision by Amish. On 2026-09-25 Amish accepted all recommendations ("i accept all your recommendations, go with them across all repos"), so D1 to D7 are now decided; see PHL-DDR-002.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in PHL-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items decided (recommendation accepted by Amish, 2026-09-25).

#	Item	Recommendation adopted	Status
D1	Bicycles in the pitch	Option (a): keep the pitch and treat a bicycle variant (own battery, handlebar or seat-post mount) as a later step. The pitch and problem lines in <code>project.yaml</code> and <code>README.md</code> are unchanged, because no rewording was recommended. R16 is redefined for this build as buses and refuse trucks, with the bicycle variant deferred.	Decided by Amish, 2026-09-25: go with recommendation
D2	Upload route	Depot Wi-Fi first; LTE-M cellular as an optional add-on, costed separately. No budget change was recommended; <code>budget_usd</code> stays at \$70.	Decided by Amish, 2026-09-25: go with recommendation
D3	Mount location	Sprung-mass floor mount above the rear axle, not an axle mount.	Decided by Amish, 2026-09-25: go with recommendation
D4	Controller	ESP32-S3 board with Wi-Fi and a microSD slot.	Decided by Amish, 2026-09-25: go with recommendation
D5	IMU and GNSS classes	LSM6DSO-class IMU and u-blox M10-class GNSS, to be confirmed against datasheets.	Decided by Amish, 2026-09-25: go with recommendation
D6	Segment length and output format	100 m segments plus point events, in CSV and GeoJSON following the CityTwin open data export.	Decided by Amish, 2026-09-25: go with recommendation
D7	Privacy rule	Publish only road-segment roughness and defect clusters; keep vehicle tracks and timestamps on the operator's server; no driver scoring. To be agreed with any fleet partner and its drivers.	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that remain open.

#	Item	Status
O1	First pilot partner and city. The TRL 2 note described the kind of partner needed (a city road department with a bus or refuse fleet that can provide reference IRI sections, possibly found through the Helpful Engineering network) but named no partner, city or preference. No choice is made here.	Decided by Amish, 2026-10-02, as recommended in PHL-DEC-001: first candidate to approach a Dallas-Fort Worth area city with its own bus or refuse fleet, with TxDOT pavement management roughness data on state roads in that city as the reference IRI sections

■ Consequences

- `project.yaml` : only the TRL fields change. No new budget, pitch or problem wording was recommended, so `budget_usd` stays at \$70 and the pitch still names bikes.
- PHL-PRB-001, PHL-PRC-001 and PHL-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed"; they are decided (PHL-DDR-002). R16 is redefined (D1); R8, R13 and R14 now rest on adopted choices rather than proposals.
- The TRL 3 calculations (PHL-CAL-001) raise two new items that this record does not decide: the converter input rating for 24 V vehicles (R9), which would take the parts cost to \$71 against the \$70 budget, and the defect-

detection speed band or reference defect (R1, R5). They were listed in docs/REVIEW.md as "Proposed, awaiting Amish" and are decided in PHL-DDR-002 (N1, N2).

- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.